Lecture 30

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1 Introduction and Motivation

1.1 Solvability of Equations

Different number systems allow for the solution of different types of equations.

- The equation $x^2 + 5 = 0$ has no solution in $\mathbb{R}$.
- The equation $\frac{2}{x} = 3$ has a solution in $\mathbb{Q}$.
- The equation $x^2 = 2$ has no solution in $\mathbb{Q}$, but it has a solution in $\mathbb{R}$.

1.2 Cardano's Formula

Consider the cubic equation $y^3 + py + q = 0$. The roots are given by the formula:

$$y = \sqrt[3]{-\frac{q}{2} + \sqrt{R}} + \sqrt[3]{-\frac{q}{2} - \sqrt{R}}$$

where $R = (\frac{p}{3})^3 + (\frac{q}{2})^2$.

1.2.1 Example

Solve $y^3 - 7y + 6 = 0$. Here $p = -7$ and $q = 6$.

$$R = \left(-\frac{7}{3}\right)^3 + 3^2 < 0$$

Despite $R < 0$, the equation has real solutions: $y_1 = 1, y_2 = 2, y_3 = -3$. This paradox (roots of negative numbers yielding real solutions) necessitates the introduction of complex numbers.

1.3 Definition

Let i be a symbol such that $i^2 = -1$. A complex number z is defined as $z = a + bi$ for $a, b \in \mathbb{R}$.

- Re $z = a$ (Real part of z)
- Im $z = b$ (Imaginary part of z)
- $|z| = \sqrt{a^2 + b^2}$ (Modulus of z)
- $\bar{z} = a - bi$ (Complex conjugate)

1.3.1 Example Calculation

Solve $x^2 - 2x + 3 = 0$.

$$D = 4 - 12 = -8$$

$$x_{1,2} = \frac{2 \pm \sqrt{-8}}{2} = \frac{2 \pm 2\sqrt{2}i}{2} = 1 \pm \sqrt{2}i$$

2 Operations

Let $z = a + bi$ and $w = c + di$.

2.1 Addition

$$z + w = (a + c) + (b + d)i$$

2.2 Multiplication

$$\begin{aligned} z \cdot w = (a + bi)(c + di) &= ac + adi + bci + bdi^2 \\ &= (ac - bd) + (ad + bc)i \end{aligned}$$

2.3 Division

$$\begin{aligned} \frac{z}{w} = \frac{a + bi}{c + di} &= \frac{(a + bi)(c - di)}{(c + di)(c - di)} \\ &= \frac{(ac + bd) + (bc - ad)i}{c^2 + d^2} \end{aligned}$$

2.4 Set Cardinality

$\mathbb{C}$ is the set of all complex numbers.

$$|\mathbb{C}| = |\mathbb{R}^2| = |\mathbb{R}| = \mathfrak{c}$$

3 Geometric Representation

Complex numbers can be represented as points in a plane (Argand plane).

- x-axis: Real axis
- y-axis: Imaginary axis

3.1 Polar / Trigonometric Form

Let $z = a + bi$.

- $a = |z| \cos\theta = r\cos\theta$
- $b = |z| \sin\theta = r\sin\theta$

$$z = r(\cos\theta + i\sin\theta)$$

where $r = |z| = \sqrt{a^2 + b^2}$.

3.2 Properties

- $z \cdot \bar{z} = |z|^2$
- $\theta = \arg z$ (Argument, measured counterclockwise).
- Multiplication by i corresponds to a $90°$ rotation. Example: $z = (1 + 2i)i = i + 2i^2 = -2 + i$.

4 Properties of Modulus

For $z, w \in \mathbb{C}$:

1. $|z \cdot w| = |z| \cdot |w|$
2. $|\frac{z}{w}| = \frac{|z|}{|w|}$
3. $|\text{Re } z| \leq |z|$
4. $|\text{Im } z| \leq |z|$
5. $|z + w| \leq |z| + |w|$ (Triangle Inequality)

4.1 Proofs

4.1.1 Proof of (1)

$$|z \cdot w|^2 = (ac - bd)^2 + (ad + bc)^2$$
$$= a^2c^2 - 2abcd + b^2d^2 + a^2d^2 + 2abcd + b^2c^2$$
$$= a^2(c^2 + d^2) + b^2(c^2 + d^2)$$
$$= (a^2 + b^2)(c^2 + d^2) = |z|^2|w|^2$$

4.1.2 Proof of (5)

We want to show $|z+w| \leq |z| + |w|$. Square both sides:

$$|z+w|^2 = (a+c)^2 + (b+d)^2 = |z|^2 + |w|^2 + 2(ac+bd)$$

$$(|z| + |w|)^2 = |z|^2 + |w|^2 + 2|z||w|$$

The inequality holds if:

$$ac + bd \leq |z||w| = \sqrt{a^2+b^2}\sqrt{c^2+d^2}$$

Squaring again (Cauchy-Schwarz context):

$$(ac+bd)^2 \leq (a^2+b^2)(c^2+d^2)$$

$$a^2c^2 + 2abcd + b^2d^2 \leq a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$$

$$2abcd \leq a^2d^2 + b^2c^2$$

$$0 \leq a^2d^2 - 2abcd + b^2c^2 = (ad-bc)^2$$

This is always true.

5 Geometric Problems

5.1 Sets in the Complex Plane

1. $\{z \in \mathbb{C} : \arg(z) = 0\}$ (Positive real axis).
2. $\{z \in \mathbb{C} : |z| \leq 1\}$ (Unit disk centered at origin).
   (a) $\{z \in \mathbb{C} : |z-i| \leq 1\}$ (Disk of radius 1 centered at i).

5.2 Geometric Meaning of Inequality

Consider $|z-1| \leq |\arg z|$ for $|z| = 1$.

- $|z-1|$ is the chord length connecting 1 and z on the unit circle.
- $|\arg z|$ is the arc length connecting 1 and z on the unit circle.
- The inequality states that the chord length is less than or equal to the arc length.